

13/7

Q) write a c++ program which includes a class "Student" find out the space occupied by object using sizeof() function.

```
#include <iostream>
#include <string>
using namespace std;
class Student {
private:
```

int id; // 4 bytes

float cgpa; // 4 bytes

char grade; // 1 byte

public:

void setData (int i, float c, char g) {

id = i;

cgpa = c;

grade = g;

}

void display() {

cout << "ID : " << id << " ,

CGPA : " << cgpa << " , Grade : "

<< grade << endl;

}

};

```
void main() {
```

```
    Student s;
```

```
    cout << "size of student object is : "  
    << sizeof(s) << "bytes" << endl;
```

```
}
```

→ According to Manual Calculation, we get $4 (\text{int}) + 4 (\text{float}) + 1 (\text{char}) = 9$ bytes. But due to Padding, compiler adds 3 extra bytes after the char to round the total size up to next multiple of 4.

This is for CPU-optimized access, which improves processing speed.

So, result: 9 bytes + 3 bytes (padding) = 12 bytes.

→ The member functions are stored in code segment of program's memory, not inside the object, so they do not add to `sizeof()` total. "padding" prevents "misaligned" memory reads.